35-Node Bayesian Network Architecture for Insider Threat Assessment

Hierarchical Causal Topology with Noisy-OR Parameterization on CMU SEI CERT R4.2 (47 Directed Edges)

LAYER 1: OBSERVABLE EVIDENCE

- AfterHoursLogin
- UnusualLoginFreq
- HighMachineCount
- WeekendLogin
- DeviceAfterHours
- AfterHoursEmail
- AfterHoursWeb
- DeviceConnectActivity
- FileCopyToRemovable
- HighFileCopyActivity
- FileAccessCount
- HighDataMovement
- ExternalEmail
- UnusualEmailVolume
- ExternalLargeEmail
- ArchiveCreation
- ExeActivity
- UnusualFileTypes
- JobSearchActivity
- CloudStorageAccess
- UnusualWebActivity

● Observable Evidence: 21 Features

LAYER 2: BEHAVIORAL INDICATORS
(9 Latent Intermediate Noisy-OR Nodes)

- **AuthAnomaly** *[Noisy-OR Latent]*
- **TimeBasedAnomaly** *[Noisy-OR Latent]*
- **RemovableMediaAnomaly** *[Noisy-OR Latent]*
- **DataAccessAnomaly** *[Noisy-OR Latent]*
- **DataMovementAnomaly** *[Noisy-OR Latent]*
- **CommunicationAnomaly** *[Noisy-OR Latent]*
- **StagingAnomaly** *[Noisy-OR Latent]*
- **FlightRiskAnomaly** *[Noisy-OR Latent]*
- **ShadowITAnomaly** *[Noisy-OR Latent]*

● Latent Behavioral Nodes: 9 Intermediate States

LAYER 3: THREAT HYPOTHESES
(4 Causal Attack Vectors)

- **UnauthorizedAccess** *[Attack Vector]*
- **DataExfiltration** *[Attack Vector]*
- **ITSabotage** *[Attack Vector]*
- **IPTheft** *[Attack Vector]*

● Threat Hypotheses: 4 Attack Classes

LAYER 4: TARGET
(Ternary Risk CPT)

- **RiskLevel** P(Low, Med, High) *[Learned CPT]*

● Target Risk Node